

CS 310 Automata Theory

Midsem: Sep 20, 2025 (16:00 - 18:00)

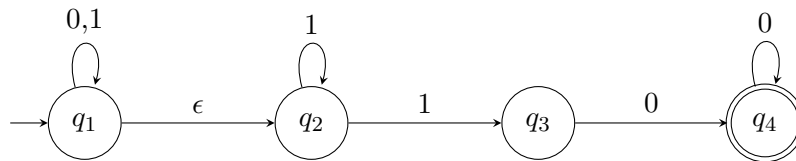
No books, calculators, mobiles, laptops.

Do rough work *first* (using **small font**) on back sides only.**Then plan** and write *concise clear* answers within the space provided. Anything outside the given box may not be evaluated. No doubts allowed.

There are 6 questions (20 marks each) on 5 pages. Fill Roll No. on all pages.

Qn. No.	1	2	3	4	5	6	Total
Marks							

1. Consider the following NFA.

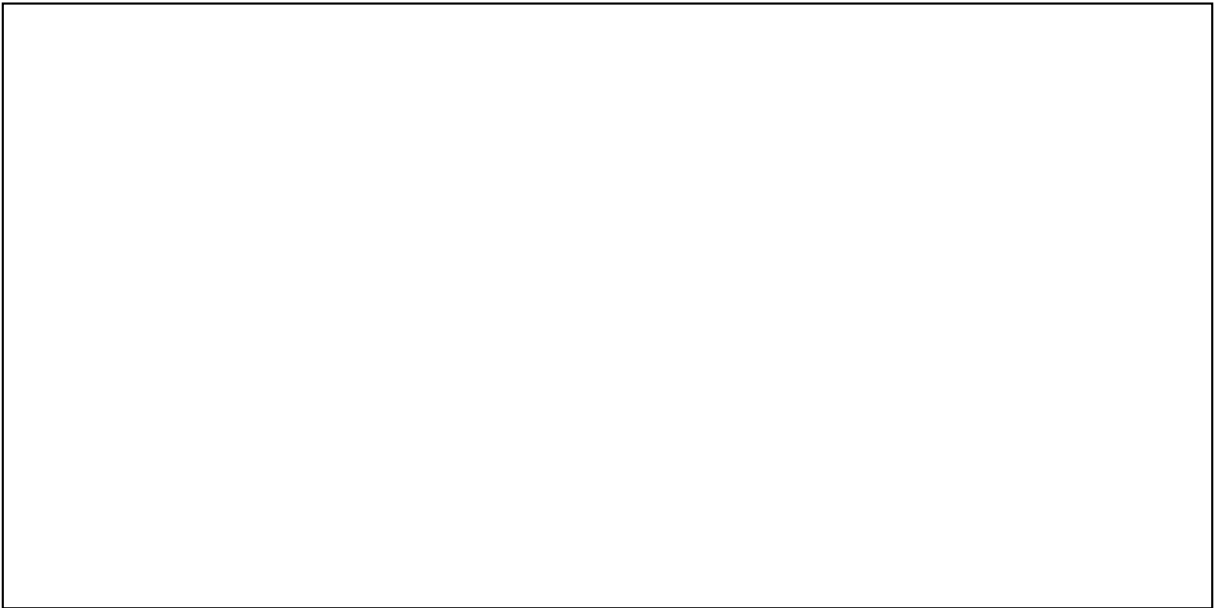
(a) Draw a **minimal** DFA accepting the language.
(b) Give a regular expression (simpler one will get more marks) R for the language L .

2. Let $L = \{w \in (0+1)^* \mid \text{if } w \text{ has } 01 \text{ as a substring then it has } 10 \text{ also as a substring}\}$
Full marks only if you give a simple/clear solution with brief justification.

(a) Construct a minimal DFA M that accepts L .



(b) Show M is minimal by giving distinguishing strings for each pair of states in M .



(c) How many strings of length 7 are there in L ?



3. Let G be the following context-free grammar.

$$\begin{aligned} S &\rightarrow ASA \mid aB \mid SC \\ A &\rightarrow B \mid S \mid \epsilon \\ B &\rightarrow b \mid \epsilon \\ C &\rightarrow cC \mid CA \mid CC \\ D &\rightarrow b \mid Da \mid Sa \end{aligned}$$

- (a) The nullable variables in G are _____.
- (b) Give (1-line each) 4 main steps needed to convert G to Chomsky normal form.

- (c) Convert G into an equivalent grammar in Chomsky Normal G_1 . Write G_1 below. More marks for less rules.

4. Let $L \subset (0+1)^*$ contain all strings that do not have equal number of 0s and 1s. Is L regular?

(a) Is L regular? **YES** **NO** (strike out wrong choice)

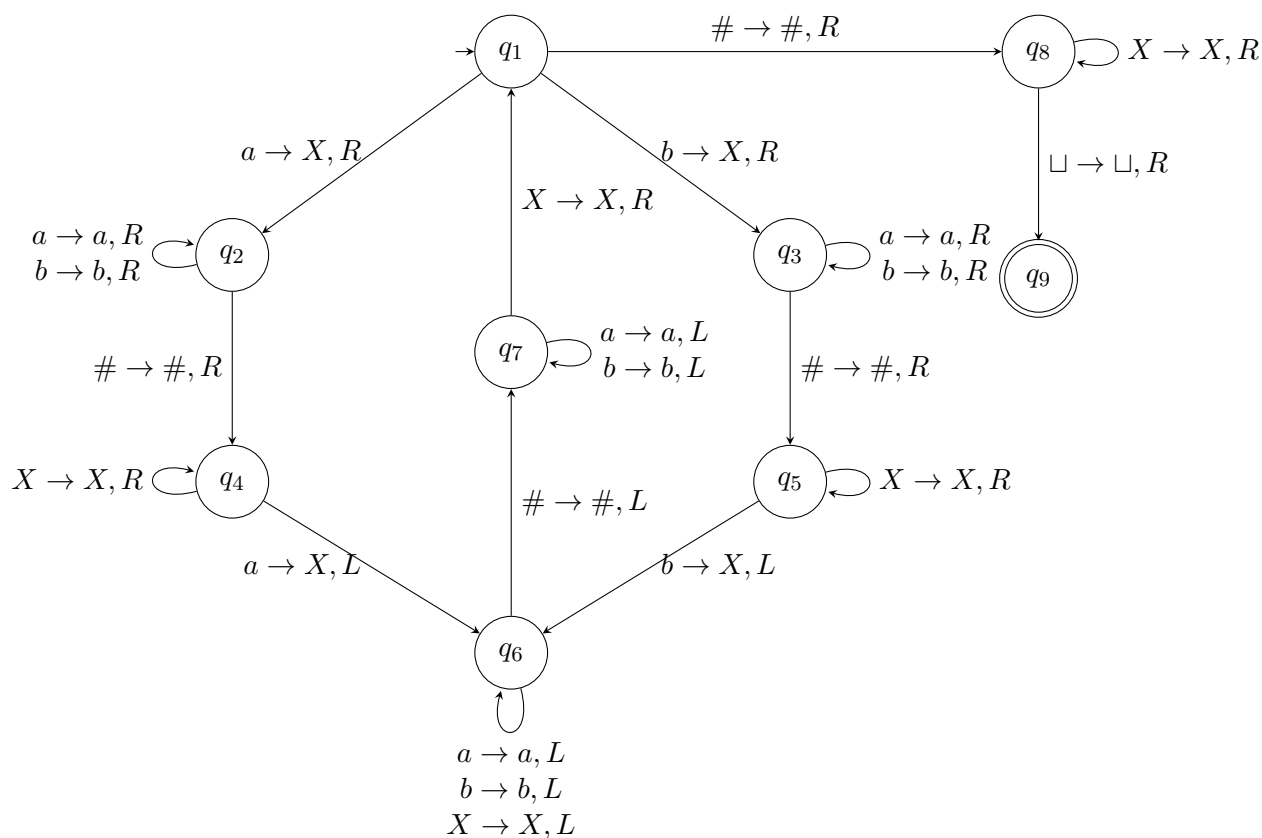
(b) Justify your choice below. Clean/crisp answers will get more marks.

5. Let $L = \{c^i b^j a^k \mid i, j, k \geq 1, i + j > k\}$

(a) Is L context-free? **YES** **NO** (strike out wrong choice)

(b) Justify your choice below. Clean/crisp answers will get more marks.

6. Consider the Turing machine M below with states $Q = \{q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9\}$ (start state q_1 , only accept state q_9) input alphabet $\Sigma = \{a, b, \#\}$, blank symbol $\sqcup$ and tape symbols $\Gamma = \Sigma \cup \{X\}$. Transitions are shown.



- (a) Assume the starting Instantaneous description (ID) of M is $q_1baab\#baba$. Show the run (sequence of IDs) below until the machine reaches accept state or crashes (rejects)

- (b) The number of strings of length 7 *not* accepted by M is ____ Give a brief justification below for your answer.